

Problems In Group Theory

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Chapter 1

Basic Group Theory

1.1 Introduction

Problem 1.1.1. Determine which of the following binary operations are associative:

1. the operation on $\mathbb{Z}$ defined by $a * b = a - b$
2. the operation on $\mathbb{R}$ defined by $a * b = a + b + ab$
3. the operation on $\mathbb{Q}$ defined by $a * b = \frac{a+b}{5}$
4. the operation on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) * (c, d) = (ad + bc, bd)$
5. the operation on $\mathbb{Q} - \{0\}$ defined by $a * b = \frac{a}{b}$

Proof. This exercise is pretty trivial as we just have to compute $a * (b * c) = (a * b) * c$. For 1., $a * (b * c) = a * (b - c) = a - (b - c) = a - b + c$ but $(a * b) * c = (a - b) * c = (a - b) - c = a - b - c$. Rest of them are pretty similar. $\square$

Problem 1.1.2. Prove that addition of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative.

Proof. By definition of residue addition,

$$[a] + [b] = [a + b]$$

Therefore, we get the following.

$$\begin{aligned} [a] + ([b] + [c]) &= [a] + [b + c] \\ &= [a + (b + c)] \\ &= [(a + b) + c] \\ &= [a + b] + [c] \\ &= ([a] + [b]) + [c] \end{aligned}$$

$\square$

Problem 1.1.3. Prove that multiplication of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative.

Proof. By definition of residue multiplication,

$$[a] \cdot [b] = [a \cdot b]$$

Therefore, we get the following.

$$\begin{aligned} [a] \cdot ([b] \cdot [c]) &= [a] \cdot [b \cdot c] \\ &= [a \cdot (b \cdot c)] \\ &= [(a \cdot b) \cdot c] \\ &= [a \cdot b] \cdot [c] \\ &= ([a] \cdot [b]) \cdot [c] \end{aligned}$$

□

Problem 1.1.4. Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.

Proof. There is no multiplicative inverse for $[a]$ with $\gcd(a, n) > 1$. The proof for it is pretty trivial, if there was an inverse, it would mean that

$$a \cdot b \equiv 1 \pmod{n} \implies 0 \equiv 1 \pmod{\gcd(a, n)}$$

That's a contradiction.

□

Problem 1.1.5. Determine which of the following sets are groups under addition:

1. the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are odd
2. the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are even
3. the set of rational numbers of absolute value < 1
4. the set of rational numbers of absolute value ≥ 1 together with 0
5. the set of rational numbers with denominators equal to 1 or 2
6. the set of rational numbers with denominators equal to 1, 2 or 3.

Proof. Below are the answers:

1. Group.
2. Not a group. Because it is not closed. (Eg: $5/2 + 7/2 = 6$)
3. Not a group. Because again it is not closed. (Eg: $0.99 + 0.98 = 1.97$)
4. Not a group. Because again it is not closed. (Eg: $4 + (-3.1) = 0.9$)
5. Group.
6. Not a group. Because it's not closed. (Eg: $1/2 + 1/3 = 5/6$)

□

Problem 1.1.6. Let $G = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$ and for $x, y \in G$ let $x * y$ be the fractional part of $x + y$ (i.e., $x * y = x + y - \lfloor x + y \rfloor$ where $\lfloor a \rfloor$ is the greatest integer less than or equal to a). Prove that $*$ is a well defined binary operation on G and that G is an abelian group under $*$ (called the real numbers mod 1).

Proof. It is well defined as,

$$\begin{aligned} x_1 = x_2, y_1 = y_2 &\implies x_1 * y_1 = x_1 + y_1 - \lfloor x_1 + y_1 \rfloor \\ &\implies x_1 + y_1 - \lfloor x_1 + y_1 \rfloor = x_2 + y_2 - \lfloor x_2 + y_2 \rfloor = x_2 * y_2 \end{aligned}$$

Also, its closed as $x * y = \{x + y\}$ and $0 \leq \{x + y\} < 1$. One can check that inverses and identity exists. For associativity,

$$\begin{aligned} x * (y * z) &= x * (y + z - \lfloor y + z \rfloor) \\ &= \{x + y + z - \lfloor y + z \rfloor\} \\ &= \{x + y + z\} \\ &= \{x + y - \lfloor x + y \rfloor + z\} \\ &= (x * y) * z \end{aligned}$$

□

Problem 1.1.7. Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$.

1. Prove that G is a group under multiplication (called the group of roots of unity in $\mathbb{C}$).
2. Prove that G is not a group under addition.

Proof. For G under multiplication, we know that for all $z_1, z_2, z_3 \in G$ we have

$$z_1 \cdot (z_2 \cdot z_3) = (z_1 \cdot z_2) \cdot z_3$$

Also, it is closed as for all $z_1, z_2 \in G$ we have $z_1^n = 1$ and $z_2^k = 1$ so

$$(z_1 \cdot z_2)^{\text{lcm}(n,k)} = 1$$

Now, for inverse of a $z \in G$ just take $1/z$. As $(1/z)^n = 1/1 = 1$ so $1/z \in G$. Also, $z \cdot 1/z = 1$.

For G under addition, if it was a group then, $1 \in G$ as $1^1 = 1 \in G \implies 1 + 1 = 2 \in G$ but there does not exist any integer n such that $2^n = 1$. □

Problem 1.1.8. Let $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$.

1. Prove that G is a group under addition.
2. Prove that the nonzero elements of G are a group under multiplication.

Proof. We know reals are associative so the elements of G are associative under addition. The inverse for $a + b\sqrt{2}$ is $(-a) + (-b)\sqrt{2}$. It is also closed as,

$$a + b\sqrt{2} + c + d\sqrt{2} = (a + c) + (b + d)\sqrt{2}$$

The identity is going to be $0 + 0 \cdot \sqrt{2} = 0$.

We know reals are associative so the elements of G are associative under multiplication. The inverse for $a + b\sqrt{2}$ is $\frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2}$. The identity is $1 + 0 \cdot \sqrt{2} = 1$. And its closed under multiplication as

$$(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$$

□

Problem 1.1.9. Prove that a finite group is abelian if and only if its group table is a symmetric matrix.

Proof. Let $G = \{g_1, \dots, g_n\}$ be a group with n elements. Then we define the group matrix to be a $n \times n$ square matrix with entries $a_{ij} = g_i g_j$. So, if its G is abelian then $a_{ij} = g_i g_j = g_j g_i = a_{ji}$, so its symmetric. If the group table is symmetric then $a_{ij} = a_{ji} \implies g_i g_j = g_j g_i$ for all $1 \leq i, j \leq n$. Thus, its abelian. □

Problem 1.1.10. Find the orders of each element of the additive group $\mathbb{Z}/12\mathbb{Z}$.

Proof. For $[1], [5], [7], [11]$ its going to be 12. For $[2], [10]$ its going to be 6, for $[4], [8]$ its going to be 3, for $[6]$ its going to be 2. For $[3], [9]$ its going to be 4. For $[0]$ its going to be 1. □

Problem 1.1.11. Prove that $(a_1 a_2 \dots a_n)^{-1} = a_n^{-1} a_{n-1}^{-1} \dots a_1^{-1}$ for all $a_1, a_2, \dots, a_n \in G$.

Proof. We will use strong induction. Its obviously true for $n = 1$ and assume its true for $n = 2, \dots, k$. Then,

$$\begin{aligned} ((a_1 \cdots a_k) \cdot a_{k+1})^{-1} &= a_{k+1}^{-1} (a_1 \cdots a_k)^{-1} \\ &= a_{k+1}^{-1} a_k^{-1} \cdots a_1^{-1} \end{aligned}$$

□

Problem 1.1.12. Let x be an element of G . Prove that $x^2 = 1$ if and only if $|x|$ is either 1 or 2.

Proof. Since $x^2 = 1$ we have $|x| \leq 2$, and since $|x|$ is a positive integer we have $|x| = 1, 2$. And if $|x| = 1$ then $x = 1 \implies x^2 = x = 1$ and if $|x| = 2$ then $x^2 = 1$. □

Problem 1.1.13. Let x be an element of G . Prove that if $|x| = n$ for some positive integer n then $x^{-1} = x^{n-1}$.

Proof. If $|x| = n$ then $x^n = 1 \implies x^{n-1} * x^1 = 1 \implies x^{n-1} = 1 * x^{-1} = x^{-1}$. □

Problem 1.1.14. For x an element in G show that x and x^{-1} have the same order.

Proof. If $|x| = \infty$ then $|x^{-1}| = \infty$ because if $|x^{-1}|$ were to be finite it would lead to a pretty obvious contradiction. If $|x| = k$ then $x^k = 1 \implies x^{-k} = 1$ so $|x^{-1}| \leq k$ and if $|x^{-1}| = i < k$ then $x^{-i} = 1 \implies x^i = 1 \implies k \leq i < k$ which is a contradiction. □

Problem 1.1.15. Let G be a finite group and let x be an element of G of order n . Prove that if n is odd, then $x = (x^2)^k$ for some k .

Proof. Since n is the order of the element x , we have $x^n = 1 \Rightarrow x^{n+1} = x$. Since $n+1 = 2k$ for some $k \in \mathbb{Z}^+$ we have

$$x^{2k} = x \Rightarrow (x^2)^k = x$$

□

Problem 1.1.16. If x and g are elements of the group G , prove that $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$.

Proof. We can easily prove through induction that $(g^{-1}xg)^k = g^{-1}x^k g$. Now let $|x| = n$ and let $|g^{-1}xg| = m$ and suppose $m < n$ then

$$g^{-1}x^m g = e \Rightarrow x^m = e \Rightarrow |x| = n \leq m < n$$

Thus, $m \geq n$ and $g^{-1}x^n g = g^{-1}eg = e$ so we have $|g^{-1}xg| = n = |x|$. Also, if you let $x = ab$ and $g = a$ we get,

$$\begin{aligned} |ab| &= |(a)^{-1}(ab)(a)| \\ &= |a^{-1}(ab)a| \\ &= |ba| \end{aligned}$$

□

Problem 1.1.17. Suppose $x \in G$ and $|x| = n < \infty$. If $n = st$ for some positive integers s and t , prove that $|x^s| = t$.

Proof. If $|x| = n$ and $n = st$ then $x^n = 1 \Rightarrow x^{st} = 1 \Rightarrow (x^s)^t = 1$. Thus, $|x^s| \leq t$. Suppose $|x^s| = k < t$ then $x^{sk} = 1$ but $n = st \leq sk \Rightarrow t \leq k < t$. Thus, $|x^s| = t$. □

Problem 1.1.18. Prove that if $x^2 = 1$ for all $x \in G$ then G is abelian.

Proof. We need to prove that $xy = yx$ for all $y, x \in G$. We know that $x^2 = y^2 = 1$ so

$$xy = x^{-1}y^{-1} = (yx)^{-1} = yx$$

as $(xy)^2 = 1$. □

Problem 1.1.19. Prove that if x is an element of the group G then $C = \{x^n \mid n \in \mathbb{Z}\}$ is a subgroup of G (called the cyclic subgroup of G generated by x).

Proof. We know the elements of C are associative because they're the element of G . The identity is going to be $x^0 = 1$ and the inverse of x^n is x^{-n} as

$$x^n \cdot x^{-n} = x^{n-n} = 1$$

Also it's closed as $x^k \cdot x^\ell = x^{k+\ell} \in C$. □

Problem 1.1.20. Let $(A, *)$ and $(B, \circ)$ be groups and let $A \times B$ be their direct product. Verify all the group axioms for $A \times B$:

1. Prove that the associative law holds: for all $(a_i, b_i) \in A \times B, i = 1, 2, 3$,
 $(a_1, b_1)[(a_2, b_2)(a_3, b_3)] = [(a_1, b_1)(a_2, b_2)](a_3, b_3),$

2. Prove that $(1, 1)$ is the identity of $A \times B$, and

3. Prove that the inverse of (a, b) is (a^{-1}, b^{-1}) .

Proof. It's just the matter of calculation. □

Problem 1.1.21. Prove that $A \times B$ is an abelian group if and only if both A and B are abelian.

Proof. Suppose $A \times B$ is abelian then,

$$\begin{aligned}(a, b)(c, d) &= (c, d)(a, b) \\ (a * c, b \circ d) &= (c * a, d \circ b)\end{aligned}$$

Thus $a * c = c * a$ and $b \circ d = d \circ b$. Therefore, both A and B are abelian. And if A and B are abelian then

$$\begin{aligned}(a, b)(c, d) &= (a * c, b \circ d) \\ &= (c * a, d \circ b) \\ &= (c, d)(a, b)\end{aligned}$$

□

Problem 1.1.22. Suppose G is a group and $|x| = n < \infty$. If $x^a = 1$ then $n \mid a$.

Proof. Using the Euclidian algorithm we get,

$$a = nq + r$$

where $0 \leq r < n$ Then,

$$x^a = x^{nq+r} = x^r = 1$$

But that's a contradiction unless $r = 0$. □

Problem 1.1.23. Prove that the elements $(a, 1)$ and $(1, b)$ of $A \times B$ commute and deduce that the order of (a, b) is the least common multiple of $|a|$ and $|b|$.

Proof. For the first statement,

$$(a, 1)(1, b) = (a * 1, 1 \circ b) = (1 * a, b \circ 1) = (1, b)(a, 1)$$

Now, suppose $|(a, b)| = n$ then,

$$\begin{aligned}(a, b)^n &= (1, 1) \\ \implies (a^n, b^n) &= (1, 1)\end{aligned}$$

Thus, $a^n = 1$ and $b^n = 1$. Therefore, $n \mid |a|$ and $n \mid |b|$. Thus, $n \mid \text{lcm}(|a|, |b|)$ which implies that $n \geq \text{lcm}(|a|, |b|)$. Now, $n = \text{lcm}(|a|, |b|)$ satisfies $(a, b)^n = (1, 1)$. □

Problem 1.1.24. Prove that any finite group G of even order contains an element of order 2.

Proof. Let $\iota(G) = \{g \in G \mid g \neq g^{-1}\}$. Then notice that order of $\iota(G)$ is even as it contains both g, g^{-1} . Then, the order of $G - \iota(G)$ is at least 2 because the order of G is even. Thus, we have a non identity element $x \in G - \iota(G)$ such that $x^2 = 1$. □

Problem 1.1.25. *If x is an element of finite order n in G , prove that the elements of*

$$C = \{x^k \mid 0 \leq k \leq n-1\}$$

are all distinct. Deduce that $|x| \leq |G|$.

Proof. Suppose that for $a \neq b$ we have,

$$x^a = x^b$$

Then, $x^{a-b} = 1$ which implies $n \mid a-b$ but $a-b < |a| < n$ which is a contradiction. Thus, all the elements of C are distinct. And since C is a subgroup we have

$$|x| = n = |C| \leq |G|$$

□